

Squashball

An analogue delay stompbox with warm, darkening repeats, self-oscillation and a STRETCH key that bends the whole echo in pitch.



Overview

Squashball is a delay pedal in the spirit of the classic bucket-brigade analogue delays. GAIN drives the signal into the delay line through a warm, slightly lopsided overdrive; the dry signal stays clean. The tone filter sits inside the feedback loop, so every repeat comes back a little darker than the last, and long delay times darken a touch more, as a slower clock does on the real thing.

REGEN goes past 100%. There the loop starts to howl into steady self-oscillation, but a soft limiter inside it holds the level, so it never runs away. Turning TIME glides the delay rather than jumping, so the repeats bend in pitch as you move it.

The STRETCH key sweeps the delay time towards a second time, STRETCH TIME, at a speed set by RATE, then glides back when you let go. Everything in the delay line bends in pitch as it goes: a fast setting sounds like a tape being yanked, a slow one

like a lazy dive. HOLD makes the key work while you hold it down; LATCH makes it click on and off.

Reach for it for

- Warm, dark analogue repeats behind a guitar or synth lead
- Rockabilly slapback and short doubling
- Dub throws and runaway, howling feedback
- Pitch-bending tape-yank and dive effects on the echoes
- Lo-fi, crunchy repeats with GAIN turned up
- Warbly chorus-like doubling at very short times

Controls

The header

PRESETS, ◀ ▶ and the display	PRESETS (or a click on the display) opens the list of factory presets; ◀ and ▶ step through them, and the display shows the one in use. A new instance opens on Init.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window. A new instance opens in Large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The main row

GAIN	Drive into the delay line, 0 to 100% (25% by default), up to about 24 dB of drive. A warm, slightly lopsided overdrive that adds even harmonics; the lows are trimmed before it so it never flubs, and the top is softened so it never fizzles. Only the repeats are driven - the dry signal stays clean.
TIME (yellow cap)	The delay time, 30 - 1500 ms (350 ms by default). With SYNC on it becomes a note value from 4 bars down to 1/32, including dotted (D) and triplet (T) values (1/4 by

	default). Turning it glides the delay, so the repeats bend in pitch rather than clicking. All times are limited to 4 seconds.
REGEN	Feedback, 0 to 130% (35% by default): how many repeats you hear. Past 100% the loop oscillates on its own and keeps going after the input stops; the limiter in the loop holds it at a steady level.
TONE	The high-frequency roll-off inside the loop, about 700 Hz - 9 kHz (0 to 100%; 55%, about 2.9 kHz, by default). The knob shows the frequency. Because it is inside the loop, each repeat passes through it again and gets darker.
LEVEL	The volume of the repeats, -24 to +6 dB (0 dB by default).
MIX	Dry / wet balance, 0 to 100% (35% by default). At 50% both are at full level; below that the repeats fade down, above it the dry signal does.

STRETCH

The settings for the STRETCH key.

TIME	The delay time the stretch sweeps towards, 30 - 2000 ms (900 ms by default). Longer than the main TIME bends the pitch down as it goes; shorter bends it up. The sweep follows an S-shaped curve, so the bend eases in and out.
RATE	How fast the stretch sweeps, 0 to 100% (50% by default): from a slow 4-second dive at 0% to a 30 ms yank at 100%. The same speed is used on the way back.
HOLD / LATCH	How the STRETCH key works. HOLD (default): the stretch lasts while you hold the key down and glides back when you let go. LATCH: click the key once to stretch and again to glide back.

The centre

Time readout	The actual delay time in ms, including any stretch and drift, so you can watch it glide.
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SYNC	Switches TIME between milliseconds (off, by default) and a note value at the project tempo (on).
TEMPO	A red light that blinks once for every repeat.
Yellow double dot	Lights while the stretch is on or still gliding back.
HOWL	Lights when REGEN is above 100% and the loop is sounding: a warning that it is self-oscillating.
STRETCH	The big lit key. Hold it (HOLD) or click it on and off (LATCH) to sweep the delay to STRETCH TIME; it lights while the stretch is on. It is also a parameter (Stretch), so you can automate it in REAPER.

IN / OUT

TRIM	Input level before anything else, -24 to +12 dB.
GUARD	Click the TRIM knob (without turning it) to switch GUARD on or off; the small light at the knob's upper right shows it is on and glows brighter while it works. Off by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 to 100% (0% by default): only the lows and highs are saturated, the mids stay clean.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). CLIP is on by default and flashes red while it clips.
DRIFT	0 to 100% (50% by default). See Drift below.
OUTPUT	Output level, -24 to +12 dB.
OUT meter	Left and right output levels.

In REAPER's parameter list only

**Bias / Sat low band /
Sat high band / Ceiling**

The rest of the output stage: the saturation's even-harmonic bias (30% by default), where its low and high bands start (40 - 400 Hz and 2 - 16 kHz) and the clip ceiling or limiter threshold (-0.3 dB by default).

Drift

DRIFT makes the delay clock breathe: a slow, small wander of the delay time - a subtle wow, a little different on the left and right, which widens the repeats - and a gentle wander in the repeats' tone. 50% is the standard amount, 100% twice that, and the wander speeds up a little as you turn it up. At 0% the delay is steady.

Knobs: drag up or down (Shift for fine moves); double-click to reset a knob to its default. The mouse wheel works too.

Presets

Presets set the main row, SYNC, the STRETCH settings and DRIFT, and switch the stretch off. OUTPUT and the IN / OUT strip are left as they are.

Init	The default settings: 350 ms, a few dark repeats, 35% mix.
Slapback	A single bright 95 ms slap with a touch of drive.
Dark Repeats	420 ms with long feedback and a very dark tone: each repeat sinks into the background.
Rubbery Lead	Synced to 1/8 D with more drive; the stretch bends towards 520 ms at a fairly slow rate for a rubbery pitch wobble.
Ambient Swell	720 ms, long feedback, the level down 1 dB and more drift; the stretch is on LATCH with a very slow dive to 1100 ms.
Self-Oscillation	REGEN at 112%: the loop howls on its own. LEVEL is down 3 dB to keep it in check.
Tape Yank	450 ms with a very fast stretch down to 60 ms: press STRETCH for a sudden tape-pull squeal.
Dub Throw	Synced to 1/4 D with long, dark feedback; the stretch

	makes a medium dive to 1200 ms.
Warble Double	A 42 ms double with full drift for a warbly, chorus-like thickening; the stretch is on LATCH.
Lo-Fi Crunch	180 ms with GAIN at 85% and a very dark tone: gritty, broken-sounding repeats.
Siren Howl	REGEN at 118% with a brighter tone; the stretch is on LATCH with a slow sweep down to 40 ms, so the howl rises like a siren.

Tips

- Try the stretch while the repeats are still sounding: it bends what is already in the line, so the trick is in the timing.
- A STRETCH TIME longer than TIME drops the pitch; shorter raises it. Set RATE low for a slow dive, high for a sudden yank.
- Use LATCH when you want to automate the stretch or play it hands-free; HOLD is quicker for one-off throws.
- For self-oscillation, set REGEN just past 100%, then ride TONE and TIME by hand. Keep LEVEL down and the output stage on CLIP or LIMIT.
- Because TONE is inside the loop, small moves make a big difference to long tails. Lower it for repeats that sit behind the part.
- Put it on a send at 100% MIX if you want to EQ or compress the repeats separately.

File: gfx-squashball.jsfx